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Energy-preserving shifted bipartite graph learning for unpaired large-scale multi-view clustering

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ABSTRACT

Anchor-based multi-view clustering has advanced significantly due to its accuracy and computational efficiency. However, it relies on the assumption that data across views are fully paired, which is often violated in practice due to asynchronous data collection. This hampers the ability to capture inter-view consistency and complementarity. Few studies have addressed this via bipartite graph alignment, but the challenges about energy disharmony and loss between bipartite graphs caused by unpaired and unbalanced anchors remain unexplored. To address this challenge, this study develops a novel framework named **Energy-preserving Shifted Bipartite Graph Learning (ESBGL)** for unpaired large-scale multi-view clustering to ensure energy harmony and preservation. Specifically, ESBGL simultaneously proposes anchor alignment learning and bipartite graph learning techniques to align anchors and bipartite graphs across views, which could avoid energy disharmony between bipartite graphs. Subsequently, a shifted bipartite graph learning paradigm is designed to preserve the cluster information and significant energy of the final consensus bipartite graph. Extensive experiments and evaluations have proved the effectiveness and superiority of our ESBGL.

1. Introduction

In recent years, with the rapid advancement of information collection technologies, the process of describing the same object from different data sources or using different feature extraction methods has generated a large amount of multi-view data (Abdullah et al., 2025a,b; Cai et al., 2022; Jiang et al., 2022, 2021; Li et al., 2025a). For instance, a person's actions can be described from multiple perspectives such as RGB images, optical flow images, and skeleton data (Hu et al., 2023; Wang et al., 2022d). Similarly, a piece of text can be disseminated in various languages, such as Chinese and English. Since most real-world data are presented in multiple views or multiple modalities, multi-view clustering methods have become increasingly important in clustering (Li et al., 2025b; Liu et al., 2025). The consistency and complementarity between multi-view data contribute to achieving more accurate clustering (Li et al., 2024b; Sun et al., 2024).

Existing multi-view clustering methods balance the consistency and complementarity between multi-view data by learning a common rep-

resentation across views, and then use this common representation to perform clustering (Fang et al., 2023; Jiang et al., 2025, 2023; Kang et al., 2020; Li et al., 2023b, 2022). The success of these clustering methods heavily relies on an assumption about the data: the data from different views are fully paired, meaning that the data describing the same object from different views are completely corresponding and sequentially arranged, as shown in Fig. 1(a). However, in practical applications, obtaining fully paired multi-view data is a challenging task due to the inevitable losses during data acquisition and transmission, as well as the inconsistency in the temporal and spatial complexities of multi-view data. As shown in Fig. 1(b), the order of the same object in different views is different, meaning that their correspondence is unknown. Multi-view clustering methods must take into account the complementary information between different views. However, the sample correspondence order in unpaired multi-view data is entirely scrambled, making it impossible to determine whether features from different views are describing the same object. This hinders multi-view clustering methods from obtaining complementary information between different

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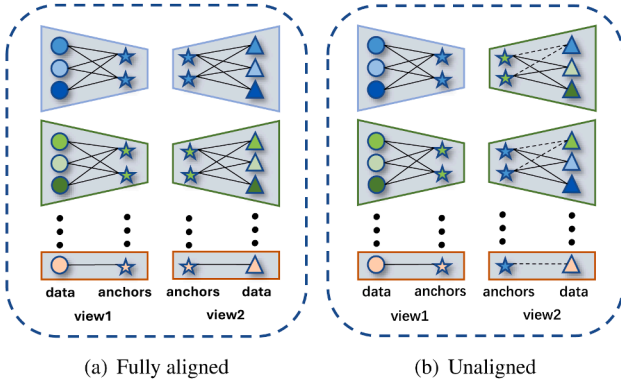


Fig. 1. Mapping Relationships Between Multi-View Data and Anchors. Solid lines represent correct mapping relationships, while dashed lines represent incorrect mapping relationships. The same color indicates the same object. Circles denote samples in View 1, triangles denote samples in View 2, and stars denote anchors.

views. Consequently, these existing works may fail when the data pairing assumption is not met.

$$\mathbf{Z} = \begin{bmatrix} \mathbf{0} & \hat{\mathbf{A}} \\ \hat{\mathbf{A}}^\top & \mathbf{0} \end{bmatrix} \in \mathbb{R}^{(n+m) \times (n+m)}.$$

Recently, several works have attempted to address this challenging issue (Lin et al., 2022b; Ren et al., 2024; Wang et al., 2022c; Yang et al., 2023b; Yu et al., 2021). However, these methods have computational complexities of $O(n^2)$ or higher, which reduces their efficiency when handling large-scale multi-view data. To reduce computational complexity, Wang et al. (2022b) proposes selecting representative anchors and aligning the bipartite graphs obtained from different views column-wise to improve clustering performance. Zhang et al. (2024) proposes a scalable framework for multi-view unsupervised feature selection by learning view-specific membership matrices and integrating structural information. However, merely aligning the bipartite graphs cannot avoid the issue of unaligned anchors learned from unpaired data. As an upstream process in the overall clustering task, unaligned anchors will reduce the final clustering performance. As shown in Fig. 1(b), the misalignment of anchors manifests in the following forms:

- Anchor unbalance problem: Anchor number of each cluster from different view are unequal.
- Anchor unfairness problem: Anchor number of different clusters from each view are unequal.
- Anchor disorder problem: Anchor order from different views are disordered.

Meanwhile, the structures between two bipartite graphs are chaotic due to anchor misalignment and sample misalignment. The chaotic graph structures would fuse into a low-quality consensus bipartite graph to degrade the clustering performance.

To address the challenges in unpaired large-scale multi-view clustering, we propose a novel approach, **Energy-preserving Shifted Bipartite Graph Learning (ESBGL)**. Specifically, to mitigate anchor misalignment, we introduce an innovative anchor-supervised learning scheme that ensures the balance, fairness, and ordering of anchors. Next, to address sample misalignment, we propose a consensus bipartite graph learning framework that generates a consistent and reliable consensus bipartite graph. Furthermore, while most existing clustering methods utilizing Laplacian matrix focus solely on the smallest eigenvalue-eigenvector pairs for encoding the primary clustering information, they overlook the significant energy embedded in the largest eigenvalue-eigenvector pairs. To overcome this limitation, we design the energy-preserving shifted learning technique, enabling the smallest eigenvalue-eigenvector pairs of the Laplacian matrix to simultaneously capture both

cluster structure and significant energy. The key contributions of this work are as follows:

- We introduces an innovative anchor-supervised learning scheme that ensures anchors are balanced, fairly distributed, and consistently ordered across different views, thereby enhancing quality of consensus bipartite graph.
- We proposes a framework to generate a consistent bipartite graph from unpaired large-scale multi-view data, capturing both cluster structure and significant energy using energy-preserving shifted learning technique.
- Extensive experiments have verified the effectiveness and high computational efficiency of our ESBGL.

2. Related work

2.1. Unaligned multi-view clustering

To the best of our knowledge, only a few works have studied the problem of unaligned Multi-View Clustering (MVC) (Yuan et al., 2025). Yu et al. (2021) explored the unknown mapping relationships between samples across views using non-negative matrix factorization. Lin et al. (2022a), Xin et al. (2024) aligned the self-representation matrices of other views to the most reliable view by selection. However, these methods have high time and space complexity, making them inefficient when handling large-scale multi-view data. Compared to shallow methods, some recent approaches are based on deep learning. Wang et al. (2022c), Xin et al. (2023), Yang et al. (2021) combined contrastive learning to enhance clustering structure and establish correspondences across views. However, these deep methods require a portion of fully paired samples as a training set. For data without paired samples, deep learning-based methods cannot cope.

2.2. Anchor-based large-scale multi-view clustering

Despite the validated performance effectiveness of existing multi-view clustering methods, their high computational complexity limits their application on large-scale multi-view datasets (Kang et al., 2020; Luo et al., 2024, 2025; Nie et al., 2017). Anchor-based multi-view clustering methods aim to guide inter-view communication through anchors to achieve more accurate and robust clustering results. By selecting or generating a small number of anchors across the entire dataset, computational complexity and memory usage can be significantly reduced, providing a feasible approach for large-scale multi-view clustering tasks. Anchor-based multi-view clustering methods can be divided into two categories: (1) Static anchor-based methods, including anchor selection methods such as random sampling and k-means (Cai & Chen, 2014; Kang et al., 2020; Li et al., 2020). (2) Dynamic anchor-based methods. These methods adaptively learn high-quality anchors and consistent bipartite graphs for clustering (Li et al., 2023c; Wang et al., 2022a; Zhang et al., 2023). The success of these methods depends on the consistence between cross-view samples. However, from Fig. 1(b), when samples across different views become unpaired, causing inconsistent bipartite graph structures across views naturally. As shown in Table 3 and Fig. 3 (a)-(d), existing methods would degrade clustering performance significantly due to unpaired samples.

3. Formulation

3.1. Motivation

Despite the success of the existing methods in addressing paired large-scale multi-view data, they would fail when facing unpaired data. First, as shown in Fig. 1(b), unpaired samples could naturally cause unaligned (i.e., cross-view number and order) and unbalanced (i.e., intra-view number) anchors due to the lack of supervisory information in the

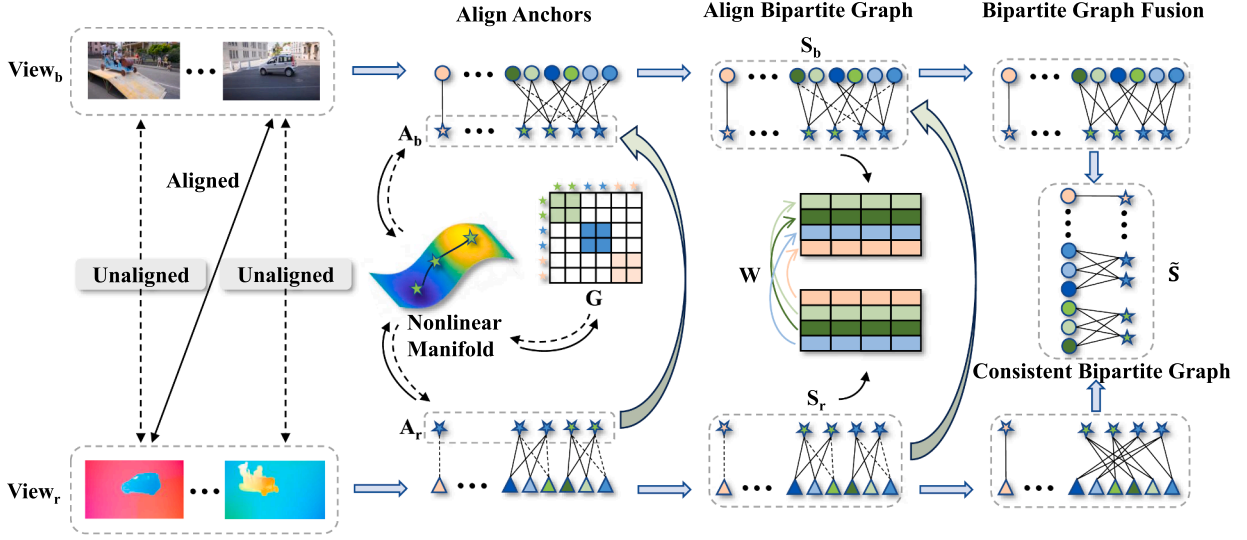


Fig. 2. Model of ESBGL including three processes: Anchor alignment learning, bipartite graph learning, and energy-preserving shifted bipartite graph learning. First, Anchor alignment learning use the prior G to learn balanced, fair, and ordered anchors by capturing local nonlinear manifold structures. Then, bipartite graph learning is designed to aligning the bipartite graphs by using the rotation matrix W to ensure consistency with the optimal bipartite graph S^b . Finally, energy-preserving shifted bipartite graph learning fuses the bipartite graphs into a consensus bipartite graph $\tilde{A}$, which guides consensus graph to extract significant energy and clustering information simultaneously.

dynamic anchor learning process. Second, from Fig. 1(b), unaligned and unbalanced anchors with unpaired samples disrupt the complementary and consensus information across views, further causing low-quality bipartite graph fusion. That is, inconsistent graph structures generated from unaligned anchors and samples cannot learn a consensus bipartite graph with a consistent structure, which causes a low-quality graph to decrease performance.

3.2. Energy-preserving shifted bipartite graph learning (ESBGL)

Considering unaligned and unbalanced anchors, we first attempt to design a novel anchored supervised learning scheme by developing a prior anchor graph to supervise the unsupervised anchor learning process, i.e.,

$$\min_{\mathbf{B}^p} \sum_{p=1}^v \sum_{i=1}^m \sum_{j=1}^m \|\mathbf{b}_i^b - \mathbf{b}_j^r\|_F^2 g_{ij} + \sum_{p=1}^v \|(\mathbf{B}^p)^T \mathbf{B}^p - \mathbf{I}_m\|_F^2 \quad (1)$$

which is simplified as $\mathcal{L}_a^p$. For the first term, the prior anchor graph $G \in \mathbb{R}^{m \times m}$ is designed to supervise the learned m cross-view anchors $\{\mathbf{B}^p\}_{p=1}^v$ to be aligned and balanced. Specifically, each g_{ij} of G encodes an affinity between two anchors of b th and r th views, where b is regarded as the highest-quality view by performing Silhouette Coefficient (Lin et al., 2022a). We set prior G as a block diagonal matrix as shown in the anchor alignment part of Fig. 2, where intra-block elements are nonzero to drag intra-cluster two anchors closer (Setting $g_{ij} = 1/m$ to protect the balance of anchors in this paper.). While inter-block elements are set to zeros to push inter-cluster two anchors away as much as possible. Thus, the learned anchor matrices tend also to be block diagonal, i.e., the learned anchors for different views would be aligned and balanced. The first term could also guide the anchors to capture the local manifold information, which plays a significant role in unsupervised learning (Nie et al., 2016b). The second term could enhance the discriminability of intra-cluster anchors since we expect the anchors of the same clusters to have better representational ability.

Second, with the aligned and balanced anchors at hand, we aim to learn a consensus bipartite graph with a consistent structure by aligning multiple bipartite graphs of different structures caused by unpaired samples. To do so, we align multiple bipartite graphs to the best-view A^b via multiple rotation matrices $\mathbf{W}^p \in \mathbb{R}^{m \times m}$, i.e., $\mathcal{L}_g^p = \|\mathbf{W}^p \mathbf{A}^p - \mathbf{A}^b\|_F^2$. In

this paper, we take anchor-based subspace learning (Wang et al., 2022b) as an example to further integrate $\mathcal{L}_a^p$ and graph matching term $\mathcal{L}_g^p$ as

$$\min_{\mathbf{A}^p, \mathbf{W}^p, \mathbf{B}^p, \mathbf{P}^p} \gamma \sum_{p=1}^v \|(\mathbf{P}^p)^T \mathbf{X}^p - \mathbf{B}^p \mathbf{A}^p\|_F^2 + \beta \mathcal{L}_a^p + \mathcal{L}_g^p \quad (2)$$

$$\text{s.t. } \mathbf{A}^p \geq 0, (\mathbf{A}^p)^T \mathbf{1} = \mathbf{1}, (\mathbf{P}^p)^T \mathbf{P}^p = \mathbf{I}_m$$

where r th anchor matrix is generated from the projection matrix $\mathbf{P}^p \in \mathbb{R}^{d^p \times k}$ with orthogonal constraint and unpaired data $\mathbf{X}^p \in \mathbb{R}^{d^p \times n}$. Constraint on $\mathbf{A}^p$ could ensure the probability attributes of graphs. Considering the different quality between views (Wen et al., 2024), the final consensus bipartite graph is weighted as $\tilde{A} = A^b + \sum_{p=1, p \neq b}^v \omega^p A^p$, where ω^p is obtained from the reconstruction loss $\mathcal{L}_g^p$ of each view, i.e., $\omega^p = 1/\mathcal{L}_g^p$. Then, the full adjacency matrix of the bipartite graph is obtained by $\mathbf{Z} = \begin{bmatrix} \mathbf{0} & \tilde{A} \\ \tilde{A}^T & \mathbf{0} \end{bmatrix} \in \mathbb{R}^{(n+m) \times (n+m)}$. Results are learned by decomposing the Laplacian matrix $\mathbf{L}_Z = \mathbf{I} - \mathbf{D}^{-\frac{1}{2}} \mathbf{Z} \mathbf{D}^{-\frac{1}{2}} \in \mathbb{R}^{(n+m) \times (n+m)}$. $\mathbf{L}_Z = \mathbf{D} - \mathbf{Z}$ where degree matrix $\mathbf{D}$ is constructed from $\mathbf{Z}$. Regrettably, directly decomposing Laplacian matrix via $\min_{\mathbf{F}} \text{Tr}(\mathbf{F}^T \mathbf{L}_Z \mathbf{F})$ only learn m smallest eigenvalue-eigenvector pairs to encode the main clustering information (You et al., 2023), while the other main energy embedding in m largest eigenvalue-eigenvector pairs are ignored. To simultaneously capture the significant energy of the largest eigenvalue-eigenvector pairs and the main clustering information of the smallest pairs, we construct the shifted Laplacian via $\tilde{\mathbf{L}}_Z = 2\mathbf{I} - \mathbf{L}_Z = \mathbf{I} + \mathbf{D}^{-\frac{1}{2}} \mathbf{Z} \mathbf{D}^{-\frac{1}{2}}$. The following theorem shows that the large eigenvalues of shifted Laplacian matrix $\tilde{\mathbf{L}}_Z$ encodes and preserve the cluster information and significant energy simultaneously (Khan & Maji, 2019).

Theorem 1. If (δ, u) is an eigenvalue-eigenvector pair of normalized Laplacian $\mathbf{L}_Z$, then $(2 - \delta, u)$ is an eigenvalue-eigenvector pair of shifted Laplacian $\tilde{\mathbf{L}}_Z$. Thus, shifted Laplacian eigenvalues lie in $[0, 2]$ (Dhanjal & Cléménçon, 2014).

Inspired by Theorem 1, we construct the shifted Laplacian for dynamic A^b upgrading rather than static $\tilde{\mathbf{L}}_Z$. Thus, by integrating Eq. (2), the final objective function becomes to

$$\min_{\mathbf{A}^p, \mathbf{W}^p, \mathbf{B}^p, \mathbf{P}^p, \mathbf{F}} \gamma \sum_{p=1}^v \|(\mathbf{P}^p)^T \mathbf{X}^p - \mathbf{B}^p \mathbf{A}^p\|_F^2 + \beta \mathcal{L}_a^p + \mathcal{L}_g^p - \lambda \text{Tr}(\mathbf{F}^T \mathbf{D}^{-\frac{1}{2}} \mathbf{Z} \mathbf{D}^{-\frac{1}{2}} \mathbf{F}) \quad (3)$$

$$\text{s.t. } \mathbf{A}^p \geq 0, (\mathbf{A}^p)^T \mathbf{1} = \mathbf{1}, (\mathbf{P}^p)^T \mathbf{P}^p = \mathbf{I}_m$$

where λ is a hidden parameter, and adaptively updated according to Ren et al. (2020). Overall, Eq. (3) proposes a new paradigm to learn aligned anchors and bipartite graphs from the unpaired multi-view data. Then, a consensus bipartite graph embedding main energy and clustering information could be learned to perform large-scale unpaired multi-view clustering.

3.3. Optimization of ESBGL

To solve Eq. (2), rewrite Eq. (2) as the following augmented Lagrangian function

$$\begin{aligned} \min_{\mathbf{A}^p, \mathbf{W}^p, \mathbf{B}^p, \mathbf{P}^p, \mathbf{F}} \gamma \sum_{p=1}^v \|\mathbf{P}^p\|_F^2 - \beta \|\mathbf{A}^p\|_F^2 + \|\mathbf{W}^p \mathbf{A}^p - \mathbf{A}^b\|_F^2 \\ + \beta \sum_{p=1}^v \sum_{i=1}^m \sum_{j=1}^m \|\mathbf{b}_i^b - \mathbf{b}_j^p\|_{F, g_{ij}}^2 - \lambda \text{Tr}(\mathbf{F}^T \mathbf{D}^{-\frac{1}{2}} \mathbf{Z} \mathbf{D}^{-\frac{1}{2}} \mathbf{F}) \end{aligned} \quad (4)$$

s.t. $\mathbf{A}^p \geq 0, (\mathbf{A}^p)^T \mathbf{1} = \mathbf{1}, (\mathbf{P}^p)^T \mathbf{P}^p = \mathbf{I}_m, (\mathbf{B}^p)^T \mathbf{B}^p = \mathbf{I}_m$

In Eq. (1), the second term allows analytical solving of $\mathbf{B}^p$, facilitating optimization. Thus, it is imposed as a constraint, and Eq. (4) is solved separately as follows:

▷ **Step-1.1: Solving $\mathbf{A}^p$ with $\mathbf{B}, \mathbf{P}, \mathbf{W}$, and $\mathbf{A}^b$.** The $\mathbf{A}^p$ -subproblem in Eq. (4) can be reformulated as:

$$\min_{\mathbf{A}^p} \|\mathbf{A}^p - \tilde{\mathbf{A}}^p\|_F^2 \text{ s.t. } \mathbf{A}^p \geq 0, (\mathbf{A}^p)^T \mathbf{1} = \mathbf{1}, \quad (5)$$

where $\tilde{\mathbf{A}}^p = \frac{(\mathbf{B}^p)^T (\mathbf{P}^p)^T \mathbf{X}^p + (\mathbf{W}^p)^T \mathbf{A}^b}{(1+\gamma)\mathbf{I}}$. Let $\tilde{\mathbf{a}}_j^p$ denote the j th column vector of $\tilde{\mathbf{A}}^p$, and its i th entry be represented as $\tilde{a}_{i,j}^p$. From the r th perspective, $\tilde{\mathbf{a}}_j^p$ and $\tilde{a}_{i,j}^p$ correspond to the j th column and the ij th element of $\mathbf{a}^p$, respectively. Consequently, the optimization of $\mathbf{a}^p$ can be formulated as a column-wise update:

$$\min_{\mathbf{a}_j^p} \|\mathbf{a}_j - \tilde{\mathbf{a}}_j^p\|_F^2, \text{ s.t. } \forall ij, \mathbf{a}_j^T \mathbf{1} = 1, \mathbf{a}_j \geq 0 \quad (6)$$

Applying Theorem 2, Eq. (6) becomes efficiently solvable.

Theorem 2. Li et al. (2023c) Given an arbitrary set of v vectors $\{\tilde{\mathbf{a}}_j\}_{j=1}^v$, closed-form optimal solution $\mathbf{a}_j^*$ is derived as

$$\mathbf{a}_j^* = \arg \min_{\mathbf{a}_j} \|\mathbf{a}_j - \tilde{\mathbf{a}}_j^p\|_F^2, \text{ s.t. } \mathbf{a}_j^T \mathbf{1} = 1, \mathbf{a}_j \geq 0 \quad (7)$$

▷ **Step-1.2: Solving $\mathbf{A}^b$ with $\mathbf{B}, \mathbf{P}, \mathbf{W}$, and $\mathbf{A}^p$.** The $\mathbf{A}^b$ -subproblem in Eq. (4) is expressed as:

$$\min_{\mathbf{A}^b} \|\mathbf{A}^b - \tilde{\mathbf{A}}^b\|_F^2 \text{ s.t. } \mathbf{A}^b \geq 0, (\mathbf{A}^b)^T \mathbf{1} = \mathbf{1}, \quad (8)$$

where $\tilde{\mathbf{A}}^b = \frac{(\mathbf{B}^b)^T (\mathbf{P}^b)^T \mathbf{X}^b + \sum_{p=1, p \neq b}^v \mathbf{W}^p \mathbf{A}^p}{(1+\gamma)\mathbf{I}}$ and $\mathbf{T}^b = \mathbf{L}_G^T \sum_{p=1}^v \mathbf{W}^p \mathbf{A}^p$. This problem can be solved analogously to $\mathbf{A}^p$ using Theorem 2.

▷ **Update-2: Solving $\mathbf{P}$ with fixed $\mathbf{A}, \mathbf{B}$, and $\mathbf{W}$.** Under this condition, the $\mathbf{P}$ -subproblem in Eq. (4) takes the form:

$$\max_{\mathbf{P}^p} \text{Tr}((\mathbf{P}^p)^T \mathbf{E}^p) \text{ s.t. } \mathbf{P}^p (\mathbf{P}^p)^T = \mathbf{I}_k, \quad (9)$$

where $\mathbf{E}^p = \mathbf{X}^p (\mathbf{A}^p)^T (\mathbf{B}^p)^T$. The optimal $\mathbf{P}^p$ can be computed via the SVD operator (Li et al., 2024a, 2023a), with per-iteration complexity of $\mathcal{O}(v\tilde{d}(nm + k^2 + km))$, where $\tilde{d} = \sum_{p=1}^v d^p$.

▷ **Step-3-1 update $\mathbf{B}^p$:** Given fixed irrelevant variables, optimizing $\mathbf{B}^p$ reduces to:

$$\max_{\mathbf{B}^p} \text{Tr}((\mathbf{B}^p)^T \mathbf{C}^p) \text{ s.t. } (\mathbf{B}^p)^T \mathbf{B}^p = \mathbf{I}_m \quad (10)$$

where $\mathbf{C}^p = \gamma (\mathbf{P}^p)^T \mathbf{X}^p (\mathbf{A}^p)^T - \beta \mathbf{B}^b \mathbf{L}_G$. The optimal $\mathbf{B}^p$ can be efficiently obtained via singular value decomposition (SVD) on $\mathbf{C}^p$.

▷ **Step-3-2 update $\mathbf{B}^b$:** Similar to $\mathbf{B}^p$, $\mathbf{B}^b$ is optimized via SVD on:

$$\mathbf{C}^b = \gamma (\mathbf{P}^b)^T \mathbf{X}^b (\mathbf{A}^b)^T - \beta \sum_{p=1, p \neq b}^v \mathbf{L}_G^T \mathbf{B}^p. \quad (11)$$

The computational complexity per iteration is $\mathcal{O}(v(ndk + nmk + m^2k))$, where $d_{all} = \sum_{p=1}^v d^p$.

▷ **Step-4 update $\mathbf{W}$:** Fixing other variables, $\mathbf{W}$ is updated as:

$$\min_{\mathbf{W}^p} \sum_{p=1}^v \|\mathbf{W}^p \mathbf{A}^p - \mathbf{A}^b\|_F^2 \text{ s.t. } (\mathbf{W}^p)^T \mathbf{W}^p = \mathbf{I}_m \quad (12)$$

where $\mathbf{A}^p = \mathbf{A}^b (\mathbf{A}^p)^T$. The solution is derived via SVD on $\mathbf{A}^p$, with per-iteration complexity $\mathcal{O}(vd(nm + km + k^2))$.

▷ **Step-5 update $\mathbf{F}$:** Solving $\mathbf{F}$ with the irrelevant variables fixed as

$$\max_{\mathbf{F}^T \mathbf{F} = \mathbf{I}} \text{Tr}(\mathbf{F}^T \mathbf{D}^{-\frac{1}{2}} \mathbf{Z} \mathbf{D}^{-\frac{1}{2}} \mathbf{F}) \quad (13)$$

Let $\mathbf{F} \in \mathbb{R}^{n \times m}$ be partitioned into $\mathbf{F}_U \in \mathbb{R}^{n \times c}$ and $\mathbf{F}_R \in \mathbb{R}^{m \times c}$, and $\mathbf{D} \in \mathbb{R}^{m \times (n+m)}$ into $\mathbf{D}_U \in \mathbb{R}^{n \times m}$ and $\mathbf{D}_R \in \mathbb{R}^{m \times m}$ respectively, leading to a further breakdown. After substituting these components, the equation rewrites to:

$$\begin{aligned} \mathbf{F}^* = \arg \max_{\mathbf{F}} 2\text{Tr}(\mathbf{F}_U^T \mathbf{K} \mathbf{F}_R) \\ \text{s.t. } \mathbf{F}_U^T \mathbf{F}_U + \mathbf{F}_R^T \mathbf{F}_R = \mathbf{I} \end{aligned} \quad (14)$$

where $\mathbf{K} \in \mathbb{R}^{n \times m}$ is defined as:

$$\mathbf{K} = \mathbf{A}^b \mathbf{D}_R \quad (15)$$

the optimal solution $\mathbf{F}^*$ is efficiently derived by the leading singular vectors of $\mathbf{K}$ in Theorem 3.

$$\max_{\mathbf{F}_U^T \mathbf{F}_U + \mathbf{F}_R^T \mathbf{F}_R = \mathbf{I}} \text{Tr}(\mathbf{F}_U^T \mathbf{K} \mathbf{F}_R) \quad (16)$$

Theorem 3. The optimal solution to Eq. (16) are:

$$\mathbf{F}_U = \frac{\sqrt{2}}{2} \mathbf{U}_1, \quad \mathbf{F}_R = \frac{\sqrt{2}}{2} \mathbf{V}_1 \quad (17)$$

where $\mathbf{U}_1$ and $\mathbf{V}_1$ are c principal left and right singular vectors of $\mathbf{K}$, respectively.

Algorithm 1 summarizes the optimization of ESBGL, with its convergence theoretically proven in the supplementary material.

Algorithm 1 The algorithm of ESBGL.

Require: Unpaired $\{\mathbf{X}^p\}_{p=1}^v$, cluster number c , latent space dimension k , β , and γ .

Set $\mathbf{P}^p = \mathbf{I}_k$ and initialize all other matrices to $\mathbf{0}$.

1: **repeat**

2: Update $\mathbf{A}^p$ and $\mathbf{A}^b$ by using Eqs. (5) and (8);

3: Update $\mathbf{P}^p, \mathbf{B}^p, \mathbf{B}^b, \mathbf{W}^p$, and $\mathbf{F}$ via Eqs (9)–(13);

4: **until** Achieve the c clusters based on the connectivity of $\mathbf{A}^b$.

Ensure: Obtain the final clustering results from $\tilde{\mathbf{A}} = \mathbf{A}^b + \sum_{p=1, p \neq b}^v \omega^p \mathbf{A}^p$.

3.4. Complexity analysis

Our ESBGL approximates to $\mathcal{O}(t(v\tilde{d}nm + v\tilde{d}(nm + k^2 + km) + v(n\tilde{d}k + m^2k + nmk) + v^2nm + \mathcal{O}(vnm))$ time complexity and $(n+k)m + 3mnv + \tilde{d}(n+m)$ space complexity. Due to $k \ll n$ and $m \ll n$, both the time and space complexities of ESBGL scale linearly with the number of samples. Supplementary material provides more details.

3.5. Convergence analysis

Theoretically, the objective function specified in Eq. (4) is bounded and possesses a lower bound, facilitating the proof of convergence. Denote $\mathbf{H} = \mathbf{D}_R \mathbf{F}_R \mathbf{F}_U^T$ and omit these constraints, enabling Fig. 4 to be

restructured into:

$$\begin{aligned}
& \min_{\mathbf{P}^p, \mathbf{W}^p, \mathbf{F}} \gamma \sum_{r=1}^v \|(\mathbf{P}^p)^T \mathbf{X}^p - \mathbf{B}^p \mathbf{A}^p\|_F^2 - 2\text{Tr}(\mathbf{A}^b \mathbf{H}) \\
& + \|\mathbf{W}^p \mathbf{A}^p - \mathbf{A}^b\|_F^2 + \beta \sum_{i=1}^m \sum_{j=1}^m \|\mathbf{b}_i^b - \mathbf{b}_j^p\|_F^2 g_{ij} \\
& \Rightarrow \min_{\mathbf{P}^p, \mathbf{W}^p, \mathbf{F}} \gamma \sum_{r=1}^v \|(\mathbf{P}^p)^T \mathbf{X}^p - \mathbf{B}^p \mathbf{A}^p\|_F^2 \\
& + \|\mathbf{A}^b - (\mathbf{W}^p \mathbf{A}^p - \mathbf{H})\|_F^2 + \beta \sum_{i=1}^m \sum_{j=1}^m \|\mathbf{b}_i^b - \mathbf{b}_j^p\|_F^2 g_{ij} \geq 0
\end{aligned} \tag{18}$$

whose lower bound is proved to be zero, which guarantees the convergence of our [Algorithm 1](#).

4. Experiment

4.1. Experimental setting

Datasets. Our experiments involved eight widely used datasets, categorized as follows: (1) Five fully unpaired multi-view datasets for benchmarking against shallow methods: YTF-100, YTF-20, YTF-10, MNIST, and MSRCv1 ([Chen et al., 2022](#); [Lin et al., 2022a](#)). (2) Two partially paired multi-view datasets, Animal and RGBD ([Dai & Van Gool, 2013](#); [Xu et al., 2022a](#)), with a pairing ratio of 50%, used for comparison with deep methods. (3) One dataset, Synthetic, designated for visualization. The unpaired data was generated by randomly permuting sample alignments across different views. Detailed dataset statistics are provided in [Table 1](#).

Comparative Methods. To the best of our knowledge, there are currently only a few works attempting to handle fully unpaired data. We compared twenty State-Of-The-Art (SOTA) multi-view clustering methods to validate the effectiveness of our ESBGL. These include: (1) Eleven aligned methods: AMGL ([Nie et al., 2016a](#)), LMSC ([Zhang et al., 2020](#)), RMSC ([Xia et al., 2014](#)), LTMSC ([Zhang et al., 2015](#)), ETLMSC ([Wu et al., 2019](#)), t-SVD-MSC ([Xie et al., 2018](#)), WTNNM ([Gao et al., 2020](#)), FPMVS ([Wang et al., 2021](#)), OMSC ([Chen et al., 2022](#)), FMVACC ([Wang et al., 2022b](#)), FSMSC ([Chen et al., 2023b](#)). (2) Two shallow unaligned methods: MVC-UM ([Yu et al., 2021](#)) and UPMGC ([Wen et al., 2024](#)). (3) Seven deep partially unaligned methods: SURE ([Yang et al., 2023a](#)), DCP ([Lin et al., 2023](#)), CVCL ([Chen et al., 2023a](#)), MFLVC ([Xu et al., 2022b](#)), DSIMVC ([Tang & Liu, 2022](#)), MVCLN ([Yang et al., 2021](#)), PVC ([Huang et al., 2020](#)). Comparing with these advanced methods thoroughly validates the effectiveness of our ESBGL.

4.2. Experiment results

In the experiments, we used four mainstream metrics to evaluate clustering performance: Accuracy (ACC), Normalized Mutual Information (NMI), Purity, and F-Score. The hardware setup for running the experiments included an Intel Core i9-14900k CPU and 32GB RAM, with

Table 1

Datasets for experiment evaluation, * denotes the unpaired datasets or competitors in this paper.

Dataset	Size	Dimensionality	Views	Classes
Synthetic*	200	{2, 2}	2	4
RGBD*	1449	{2048, 300}	2	13
Animal*	10158	{4096, 4096}	2	50
YTF-100*	195537	{944, 576, 512, 640}	4	100
YTF-20*	63896	{944, 576, 512, 640}	4	20
YTF-10*	38,654	{944, 576, 512, 640}	4	10
Mnist*	2000	{9, 9, 30}	3	10
MSRCv1*	210	{24, 512, 256, 254}	4	7

Matlab2024a software on a Windows 11 operating system. To assess the effectiveness and robustness of ESBGL, we conducted 20 repeated experiments. [Table 2](#) presents the mean clustering performance and standard deviation of ESBGL on two fully unpaired datasets. [Table 3](#) reports the clustering results of four anchor-based methods on both fully paired and fully unpaired large-scale multi-view datasets. Other shallow methods were excluded due to their prohibitive time complexity $O(n^3)$ and space complexity $O(n^2)$. The best and second-best results are highlighted in bold and underlined, respectively. [Fig. 3](#) visualizes the bipartite graphs learned by FMVACC and ESBGL from the Synthetic dataset. Based on these tables and figures, we conclude the following:

Compared with shallow fully unaligned methods, the effectiveness and superiority of ESBGL are mainly reflected in two aspects.

(1) ESBGL, along with UPMGC and MVC-UM, significantly outperforms tensor-based multi-view clustering methods (LTMSC, ETLMSC, t-SVD-MSC and WTNNM), graph-based MVC methods (AMGL, RMSC and LMSC). This advantage is attributed to the carefully designed alignment module. Our ESBGL can learn high-quality anchor features in refined subspaces, reducing more redundant information compared to UPMGC and MVC-UM.

(2) ESBGL clearly outperforms anchor-based shallow MVC methods (OMSC, FPMVS, FMVACC, FSMSC). The primary reason is that, when performing fully unaligned multi-view clustering tasks, unpaired data disrupts the natural inter-view consistency and complementarity inherent in paired data. As shown in [Table 3](#), these anchor-based shallow methods exhibit high performance on paired datasets; however, their performance significantly declines on fully unaligned datasets. [Fig. 3](#) further corroborates that in an unpaired data environment, traditional anchor-based methods almost fail to learn a consistent bipartite graph structure. FMVACC attempts to align the bipartite graph but neglects the upstream task of aligning anchors, which reduces the quality of its bipartite graph, demonstrating the effectiveness and superiority of ESBGL.

4.3. Ablation studies

To evaluate the effectiveness and importance of dual alignment of ESBGL, we conducted an ablation study named w/o $\mathcal{L}_a^p$, representing the removal of the anchor alignment process, respectively. [Fig. 4](#) shows that removing this component leads to a significant decline in ESBGL's performance. Specifically, the anchor alignment process ensures that the learned anchors satisfy several key attributes, including intra-view inter-cluster discriminative, intra-view intra-cluster compressible, maximization of intra-view diversity and alignment of the number and order of anchors across views. [Fig. 3\(b\)](#) and (e) illustrate the decline in clustering performance after removing L. Figures [Fig. 3\(c\)](#) and (f) visually demonstrate the significance of L for cross-view anchor alignment. These results demonstrate that the dual alignment of anchors is crucial and effective.

4.4. More experimental results

The effectiveness and superiority of our ESBGL compared to deep methods are evident. Most deep learning-based methods can only handle partially unpaired data because they require a portion of paired data as a training set to train the model. To compare with deep methods, we set the pairing ratio of the two partially paired multi-view datasets to 50%. As shown in [Table 5](#), our ESBGL consistently outperforms all comparison methods. This further validates the effectiveness and superiority of ESBGL.

ESBGL demonstrates high efficiency. [Table 4](#) presents a comparison of the time complexity and space complexity between ESBGL, UPMGC, and MVC-UM. In terms of memory cost, both UPMGC and MVC-UM exhibit quadratic complexity, whereas ESBGL demonstrates linear complexity. Regarding time cost, UPMGC and MVC-UM show cubic complexity, while ESBGL displays linear complexity. [Fig. 5](#) reports the

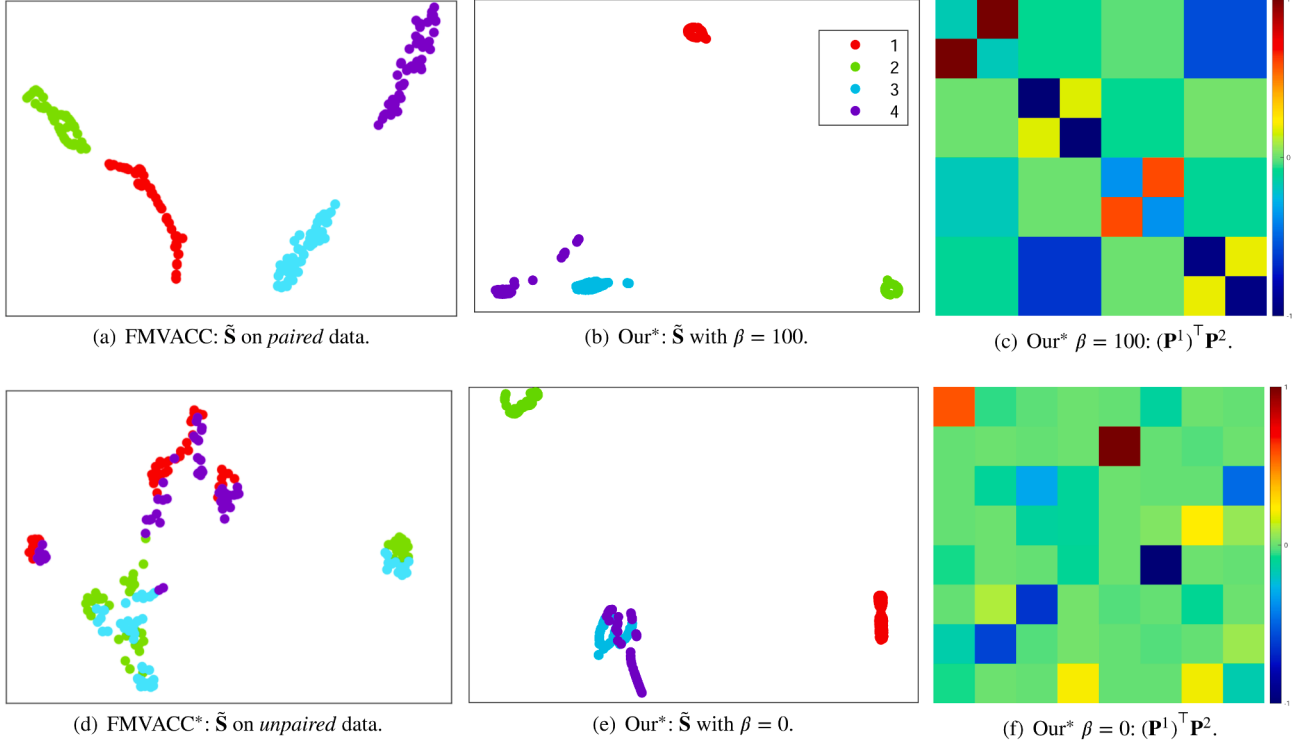


Fig. 3. (a) and (d) are the visualization of $\tilde{\mathbf{S}}$ on paired and unpaired data; (b) and (e) are the visualization of $\tilde{\mathbf{S}}$ when $\beta = 100$ and $\beta = 0$; (c) and (d) denote cross-view anchor similarities $(\mathbf{P}^1)^T \mathbf{P}^2$ when $\beta = 100$ and $\beta = 0$.

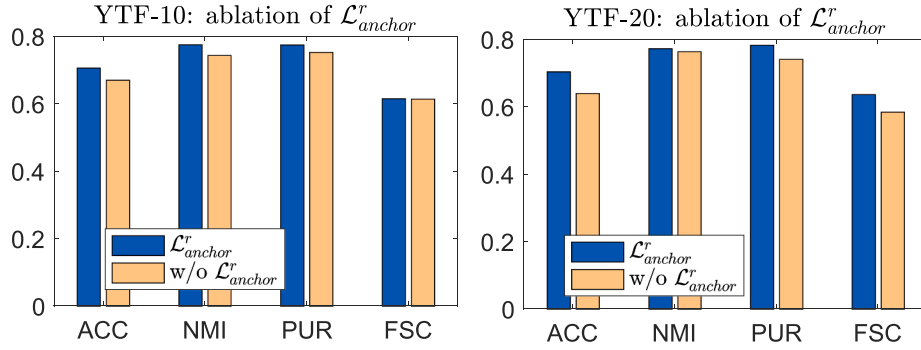


Fig. 4. The performance comparison of ablation analysis on fully unpaired large-scale datasets.

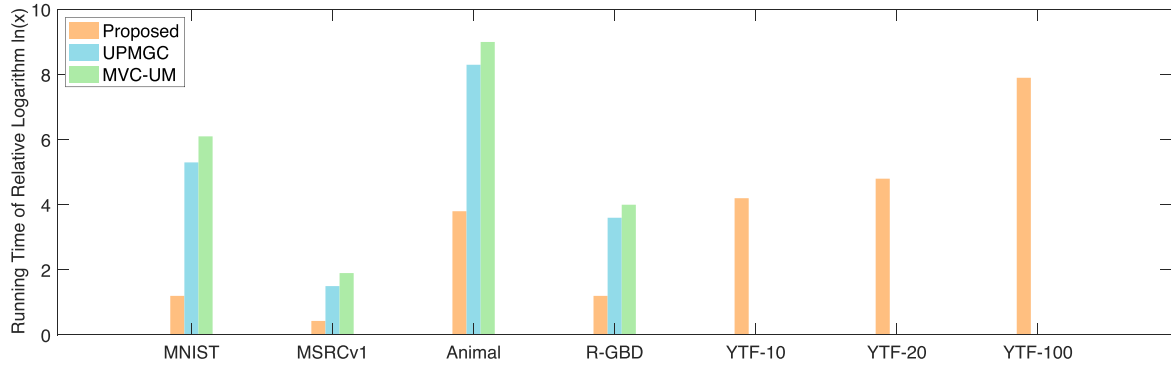


Fig. 5. Running time comparison of the most related unpaired multi-view clustering methods.

Table 2Performance comparison *w.r.t.* 13 shallow methods, where best and second-best averages are marked in bold and underlined, respectively.

Method	Mnist*				MSRCv1*			
	ACC	NMI	PUR	F-score	ACC	NMI	PUR	F-score
AMGL (IJCAI'16 Nie et al., 2016a)	17.08 $\pm$ 0.01	03.96 $\pm$ 0.00	19.73 $\pm$ 0.01	12.05 $\pm$ 0.00	29.71 $\pm$ 0.02	12.63 $\pm$ 0.02	30.38 $\pm$ 0.02	18.43 $\pm$ 0.01
LMSC (TPAMI'20 Zhang et al., 2020)	50.78 $\pm$ 0.00	48.77 $\pm$ 0.00	63.06 $\pm$ 0.00	43.24 $\pm$ 0.00	39.33 $\pm$ 0.08	25.15 $\pm$ 0.07	41.62 $\pm$ 0.06	27.03 $\pm$ 0.06
RMSC (AAAI'14 Xia et al., 2014)	42.12 $\pm$ 0.03	31.75 $\pm$ 0.02	46.79 $\pm$ 0.01	33.76 $\pm$ 0.03	37.43 $\pm$ 0.02	27.35 $\pm$ 0.04	39.52 $\pm$ 0.03	26.96 $\pm$ 0.02
LTMSC (TPAMI'15 Zhang et al., 2015)	37.54 $\pm$ 0.00	25.98 $\pm$ 0.00	40.14 $\pm$ 0.00	29.68 $\pm$ 0.00	34.57 $\pm$ 0.02	17.89 $\pm$ 0.02	36.29 $\pm$ 0.01	22.11 $\pm$ 0.01
ETLMSC (TIP'19 Wu et al., 2019)	12.82 $\pm$ 0.01	0.84 $\pm$ 0.00	16.89 $\pm$ 0.00	10.47 $\pm$ 0.00	22.95 $\pm$ 0.01	5.09 $\pm$ 0.01	23.77 $\pm$ 0.01	14.28 $\pm$ 0.01
t-SVD-MSC (IJCV'18 Xie et al., 2018)	39.47 $\pm$ 0.00	28.04 $\pm$ 0.00	41.93 $\pm$ 0.00	32.34 $\pm$ 0.00	40.76 $\pm$ 0.01	25.86 $\pm$ 0.01	42.48 $\pm$ 0.00	28.94 $\pm$ 0.01
WTNNM (AAAI'20 Gao et al., 2020)	38.70 $\pm$ 0.00	22.30 $\pm$ 0.00	42.09 $\pm$ 0.00	24.90 $\pm$ 0.00	42.67 $\pm$ 0.00	27.96 $\pm$ 0.01	44.10 $\pm$ 0.01	30.23 $\pm$ 0.01
FPMVS (TIP'22 Wang et al., 2021)	16.25 $\pm$ 0.00	2.54 $\pm$ 0.00	18.55 $\pm$ 0.00	13.73 $\pm$ 0.00	23.81 $\pm$ 0.00	7.65 $\pm$ 0.00	25.71 $\pm$ 0.00	17.15 $\pm$ 0.00
OMSC (KDD'22 Chen et al., 2022)	16.85 $\pm$ 0.00	2.44 $\pm$ 0.00	20.10 $\pm$ 0.00	12.81 $\pm$ 0.00	23.81 $\pm$ 0.00	7.26 $\pm$ 0.00	25.71 $\pm$ 0.00	17.15 $\pm$ 0.00
FMVACC (NIPS'22 Wang et al., 2022b)	29.95 $\pm$ 0.03	15.19 $\pm$ 0.03	36.90 $\pm$ 0.03	20.02 $\pm$ 0.03	37.52 $\pm$ 0.02	25.85 $\pm$ 0.02	39.72 $\pm$ 0.02	25.93 $\pm$ 0.01
FSMSC (TIP'23 Chen et al., 2023b)	18.21 $\pm$ 0.00	5.44 $\pm$ 0.00	21.17 $\pm$ 0.00	13.08 $\pm$ 0.00	28.57 $\pm$ 0.00	7.73 $\pm$ 0.00	28.57 $\pm$ 0.00	16.79 $\pm$ 0.00
MVC-UM* (KDD'21 Yu et al., 2021)	65.40 $\pm$ 0.01	54.26 $\pm$ 0.01	68.10 $\pm$ 0.01	56.69 $\pm$ 0.01	44.96 $\pm$ 0.01	34.06 $\pm$ 0.01	47.08 $\pm$ 0.01	41.30 $\pm$ 0.01
UPMGC* (TNNLS'24 Wen et al., 2024)	<u>81.82</u> $\pm$ 0.01	<u>76.30</u> $\pm$ 0.01	<u>87.31</u> $\pm$ 0.01	<u>75.47</u> $\pm$ 0.02	<u>71.22</u> $\pm$ 0.03	<u>62.76</u> $\pm$ 0.02	<u>73.74</u> $\pm$ 0.02	<u>59.38</u> $\pm$ 0.03
ESBGL*	87.55 $\pm$ 0.00	76.21 $\pm$ 0.00	87.55 $\pm$ 0.00	78.99 $\pm$ 0.00	74.76 $\pm$ 0.01	65.19 $\pm$ 0.00	74.76 $\pm$ 0.01	60.79 $\pm$ 0.00

Table 3The performance comparison *w.r.t.* anchor based shallow methods, where best results are marked in bold.

Method	YTF100				YTF100*			
	ACC	NMI	PUR	F-score	ACC	NMI	PUR	F-score
FPMVS (TIP'22 Wang et al., 2021)	52.93 $\pm$ 0.00	75.32 $\pm$ 0.00	54.46 $\pm$ 0.00	35.41 $\pm$ 0.00	16.39 $\pm$ 0.00	1.68 $\pm$ 0.00	18.11 $\pm$ 0.00	54.78 $\pm$ 0.00
OMSC (KDD'22 Chen et al., 2022)	66.51 $\pm$ 0.01	83.37 $\pm$ 0.01	71.41 $\pm$ 0.01	58.46 $\pm$ 0.01	17.63 $\pm$ 0.00	2.09 $\pm$ 0.00	20.27 $\pm$ 0.00	12.80 $\pm$ 0.00
FMVACC (NIPS'22 Wang et al., 2022b)	63.58 $\pm$ 0.01	82.09 $\pm$ 0.01	71.78 $\pm$ 0.01	38.13 $\pm$ 0.02	41.15 $\pm$ 0.04	44.08 $\pm$ 0.05	44.69 $\pm$ 0.05	18.59 $\pm$ 0.03
FSMSC (TIP'23 Chen et al., 2023b)	62.13 $\pm$ 0.01	82.13 $\pm$ 0.01	70.68 $\pm$ 0.01	40.62 $\pm$ 0.01	10.92 $\pm$ 0.02	0.96 $\pm$ 0.01	16.05 $\pm$ 0.02	8.92 $\pm$ 0.03
ESBGL*	67.26 $\pm$ 0.00	83.54 $\pm$ 0.00	72.65 $\pm$ 0.00	58.93 $\pm$ 0.00	67.19 $\pm$ 0.00	79.45 $\pm$ 0.00	72.55 $\pm$ 0.00	56.46 $\pm$ 0.00
Method	YTF20				YTF20*			
	ACC	NMI	PUR	F-score	ACC	NMI	PUR	F-score
FPMVS (TIP'22 Wang et al., 2021)	69.48 $\pm$ 0.00	77.90 $\pm$ 0.00	72.59 $\pm$ 0.00	62.61 $\pm$ 0.00	8.04 $\pm$ 0.00	0.56 $\pm$ 0.00	10.54 $\pm$ 0.00	7.21 $\pm$ 0.00
OMSC (KDD'22 Chen et al., 2022)	74.46 $\pm$ 0.00	81.70 $\pm$ 0.00	77.31 $\pm$ 0.00	68.35 $\pm$ 0.00	7.94 $\pm$ 0.00	0.70 $\pm$ 0.00	11.02 $\pm$ 0.00	5.87 $\pm$ 0.00
FMVACC (NIPS'22 Wang et al., 2022b)	66.83 $\pm$ 0.02	78.27 $\pm$ 0.01	73.18 $\pm$ 0.01	55.46 $\pm$ 0.05	10.42 $\pm$ 0.00	3.37 $\pm$ 0.00	13.05 $\pm$ 0.01	7.14 $\pm$ 0.00
FSMSC (TIP'23 Chen et al., 2023b)	67.93 $\pm$ 0.01	79.21 $\pm$ 0.02	74.35 $\pm$ 0.01	60.27 $\pm$ 0.01	15.36 $\pm$ 0.01	3.67 $\pm$ 0.00	18.27 $\pm$ 0.01	9.11 $\pm$ 0.00
ESBGL*	72.16 $\pm$ 0.01	77.61 $\pm$ 0.00	75.79 $\pm$ 0.01	63.82 $\pm$ 0.00	70.24 $\pm$ 0.00	77.08 $\pm$ 0.00	78.14 $\pm$ 0.00	63.53 $\pm$ 0.00
Method	YTF10				YTF10*			
	ACC	NMI	PUR	F-score	ACC	NMI	PUR	F-score
FPMVS (TIP'22 Wang et al., 2021)	73.25 $\pm$ 0.00	77.40 $\pm$ 0.00	76.21 $\pm$ 0.00	69.59 $\pm$ 0.00	13.52 $\pm$ 0.00	0.21 $\pm$ 0.00	16.21 $\pm$ 0.00	13.69 $\pm$ 0.00
OMSC (KDD'22 Chen et al., 2022)	78.20 $\pm$ 0.00	82.75 $\pm$ 0.00	82.98 $\pm$ 0.00	74.56 $\pm$ 0.00	12.07 $\pm$ 0.00	0.51 $\pm$ 0.00	16.71 $\pm$ 0.00	11.37 $\pm$ 0.00
FMVACC (NIPS'22 Wang et al., 2022b)	76.52 $\pm$ 0.01	62.70 $\pm$ 0.01	66.15 $\pm$ 0.01	70.20 $\pm$ 0.04	13.31 $\pm$ 0.01	0.59 $\pm$ 0.01	16.76 $\pm$ 0.01	11.90 $\pm$ 0.04
FSMSC (TIP'23 Chen et al., 2023b)	77.23 $\pm$ 0.01	72.13 $\pm$ 0.02	73.24 $\pm$ 0.01	72.99 $\pm$ 0.02	20.13 $\pm$ 0.01	8.21 $\pm$ 0.01	25.41 $\pm$ 0.02	12.43 $\pm$ 0.01
ESBGL*	80.37 $\pm$ 0.00	75.89 $\pm$ 0.00	80.14 $\pm$ 0.00	65.67 $\pm$ 0.00	70.50 $\pm$ 0.00	77.37 $\pm$ 0.00	77.32 $\pm$ 0.00	61.40 $\pm$ 0.00

Table 4

Space and time complexities of competitors for unaligned multi-view clustering.

Method	Memory Cost $\approx$	Time Cost $\approx$
MVC-UM* (KDD'21 Yu et al., 2021)	$\mathcal{O}(vn^2)$	$\mathcal{O}(3vn^3 + n^3)$
UPMGC* (TNNLS'24 Wen et al., 2024)	$\mathcal{O}(vn^2)$	$\mathcal{O}(3vn^3 + n^3)$
ESBGL*	$\mathcal{O}(vn)$	$\mathcal{O}(vn)$

time cost comparison between ESBGL, MVC-UM, and UPMGC. The results indicate that our ESBGL significantly reduces time costs compared

to MVC-UM and UPMGC. This result clearly confirms the efficiency advantage of ESBGL. In addition, Fig. 5 indicates that the time cost of our ESBGL is far below that of existing fully unpaired competitors, demonstrating the efficiency of ESBGL.

4.5. Parameter settings

Algorithm 1 requires proper setting of two parameters: γ and β . Throughout this work, we use and fix $m = 2c$ anchors for all datasets. The consensus dimension is also fixed at $k = 2c$. As illustrated in Fig. 6, our ESBGL exclusively provides the parameter settings and convergence for YTF20*.

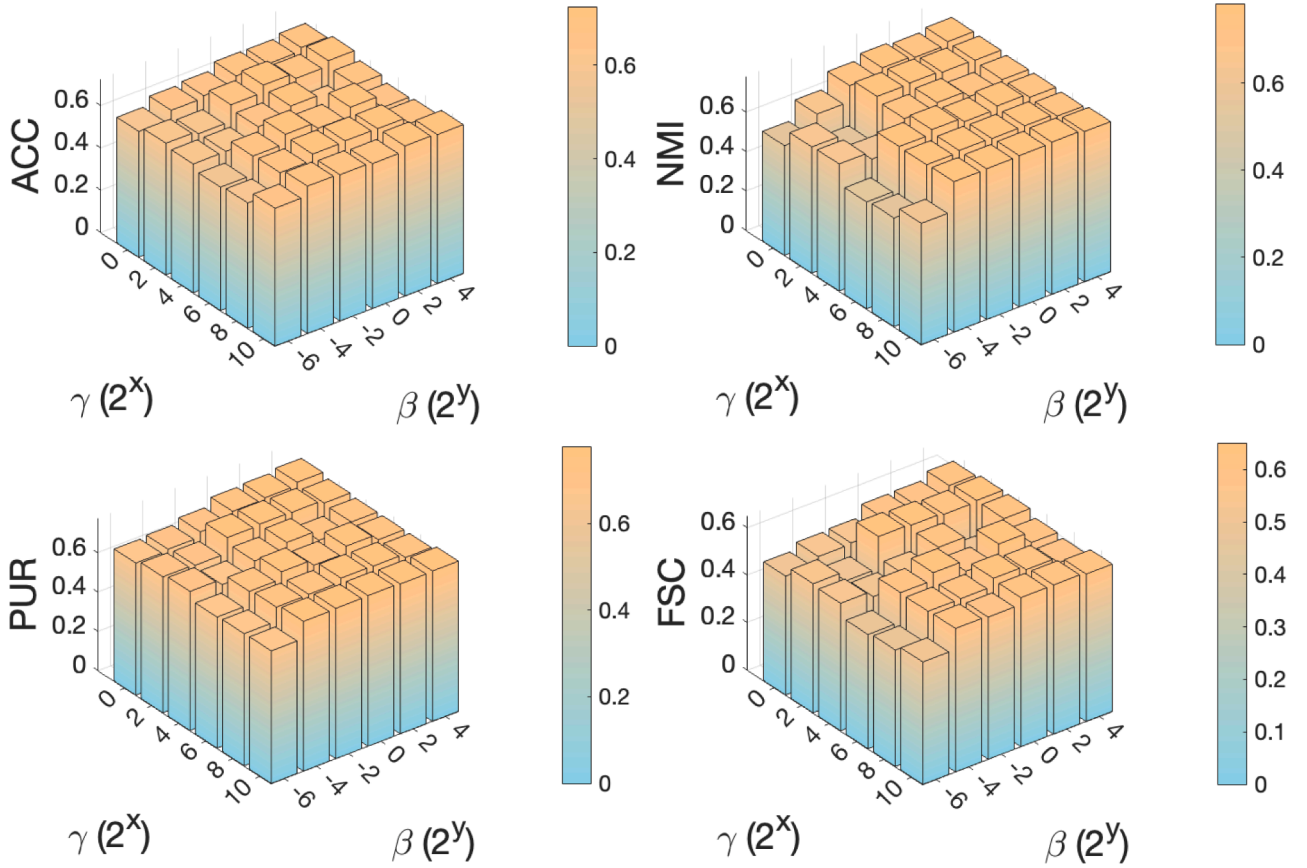


Fig. 6. The parameter settings on YTF20*.

Table 5
The performance comparison of deep methods.

Dataset	Metric	PVC	MVCLN	DSIMVC	MFLVC	CVCL	DCP	SURE	MVC-UM	UPMGC	ESBGL
Animal*	ACC	3.80	26.18	7.41	6.94	6.51	19.86	27.74	24.3	32.63	34.37
	NMI	0.10	40.19	6.26	5.09	5.79	21.46	40.83	32.16	33.91	37.16
RGBD*	ACC	18.63	25.95	28.50	25.33	14.77	25.49	35.68	28.13	32.61	43.93
	NMI	1.64	18.35	1.79	4.97	4.89	8.65	33.26	25.26	30.91	37.10

4.6. Conclusion

In this paper, we introduce an innovative ESBGL framework specifically tailored for unaligned large-scale multi-view clustering tasks. To the best of our knowledge, ESBGL represents the first effort to address the problem of unpaired multi-view data using a dual alignment technique involving both anchors and bipartite graphs. Our carefully designed anchor extraction and bipartite graph matching techniques effectively obtain aligned anchors and bipartite structures from fully unpaired multi-view data. Comprehensive experiments and analyses validate efficiency and performance of ESBGL.

CRedit authorship contribution statement

Xingfeng Li: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Data curation, Conceptualization; **Jiawei Peng:** Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization; **Zhongwen Wang:** Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodol-

ogy, Investigation, Formal analysis, Data curation, Conceptualization; **Yuan Sun:** Visualization, Software, Project administration, Methodology, Investigation, Data curation; **Yuying Zhu:** Writing – original draft, Visualization, Supervision, Methodology, Investigation, Formal analysis, Data curation, Conceptualization; **Fei Zhou:** Writing – original draft, Validation, Software, Methodology, Investigation, Data curation; **Zhenwen Ren:** Writing – review & editing, Visualization, Supervision, Software, Methodology, Funding acquisition, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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